

DWV Assignment 6 Swiston

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```
#Open the library and have it read in the data set. Make sure ggplot is activated
library(haven)
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.2.0
```

```
v forcats   1.0.1      v stringr    1.6.0
```

```
v ggplot2    4.0.3      v tibble     3.3.1
```

```
v lubridate  1.9.5      v tidyr      1.3.2
```

```
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
mydata <- read_dta("/Users/brynsuiston/Documents/Data Wrangling Library/Week 6 Homework Files
```

```
library (ggplot2)
```

```
#Separate out the 15 variables of interest, and then turn them into a distinct data set  
my_15_vars <- c("year", "spkath", "colath", "libath", "spkrac", "colrac", "librac", "spkcom"
```

```
filtered_data <- mydata[, my_15_vars]
```

```
#Filter the data by years of interest  
new_filtered_data <- filtered_data |> filter(year %in% 1976:1998)
```

```
#Check to make sure data was filtered correctly  
table(new_filtered_data$spkath)
```

```
      1      2  
15065  6613
```

```
table(new_filtered_data$colath)
```

```
      4      5  
10542 10557
```

```
table(new_filtered_data$libath)
```

```
      1      2  
7185 14113
```

```
#Recode the new filtered data as either tolerant or intolerant  
new_filtered_data <- new_filtered_data |>  
  mutate(  
  
    spkath_r = case_when(spkath == 2 ~ 1, spkath == 1 ~ 0),  
    spkrac_r = case_when(spkrac == 2 ~ 1, spkrac == 1 ~ 0),  
    spkcom_r = case_when(spkcom == 2 ~ 1, spkcom == 1 ~ 0),
```

```

spkmil_r = case_when(spkmil == 2 ~ 1, spkmil == 1 ~ 0),
spkhomo_r = case_when(spkhomo == 2 ~ 1, spkhomo == 1 ~ 0),

colath_r = case_when(colath == 5 ~ 1, colath == 4 ~ 0),
colrac_r = case_when(colrac == 5 ~ 1, colrac == 4 ~ 0),
colmil_r = case_when(colmil == 5 ~ 1, colmil == 4 ~ 0),
colhomo_r = case_when(colhomo == 5 ~ 1, colhomo == 4 ~ 0),

colcom_r = case_when(colcom == 4 ~ 1, colcom == 5 ~ 0),

libath_r = case_when(libath == 1 ~ 1, libath == 2 ~ 0),
librac_r = case_when(librac == 1 ~ 1, librac == 2 ~ 0),
libcom_r = case_when(libcom == 1 ~ 1, libcom == 2 ~ 0),
libmil_r = case_when(libmil == 1 ~ 1, libmil == 2 ~ 0),
libhomo_r = case_when(libhomo == 1 ~ 1, libhomo == 2 ~ 0)
)

```

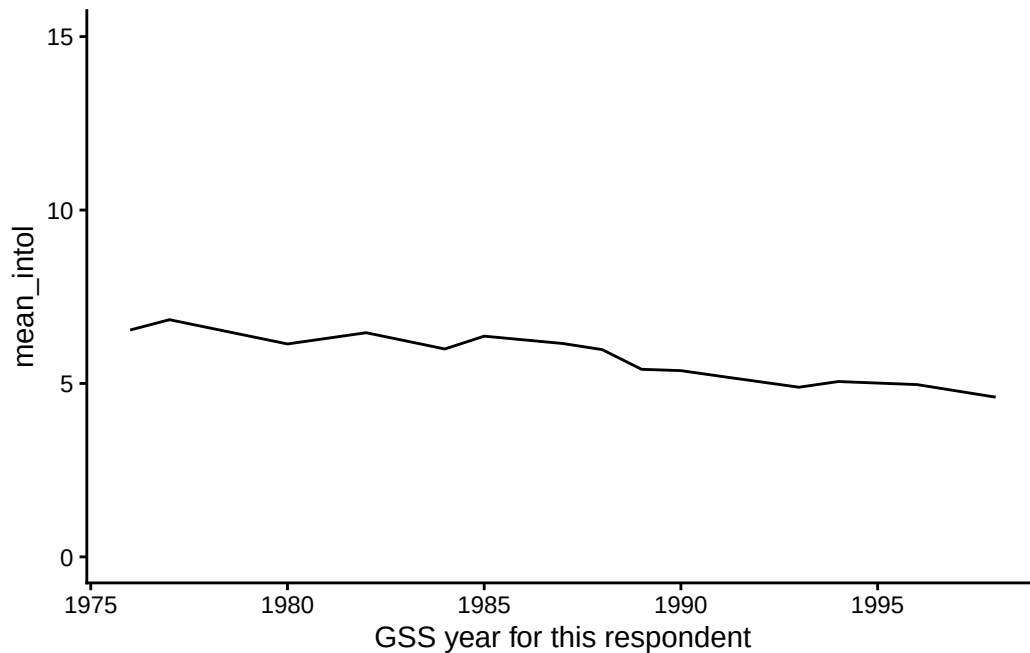
```

#Regroup the recoded and filtered data by year. Create a line graph
new_filtered_data <- new_filtered_data |>
  mutate(intol = rowSums(across(ends_with("_r")))) |>
  filter(!is.na(intol))

yearly <- new_filtered_data |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 15)) +
  theme_classic()

```



```
#Separate out the six variables of interest
my_6_vars <- c("year", "spkrac", "colrac", "librac", "spkhomo", "colhomo", "libhomo")
filtered_data2<- mydata[, my_6_vars]
```

```
#Separate out racist
my_rac_vars <- c("year", "spkrac", "colrac", "librac")
filtered_data2rac <- mydata[, my_rac_vars]
```

```
#Separate homosexuality
my_homo_vars <- c("year", "spkhomo", "colhomo", "libhomo")
filtered_data2homo <- mydata[, my_homo_vars]
```

```
#Filter racist by year
new_filtered_data2rac <- filtered_data2rac |> filter(year %in% 1976:1998)
```

```
#Filter homosexuality by year
new_filtered_data2homo <- filtered_data2homo |> filter(year %in% 1976:1998)
```

```
#Recode the racist data as either tolerant or intolerant
new_filtered_data2rac <- new_filtered_data2rac |>
  mutate(
```

```

    spkrac_r = case_when(spkrac == 2 ~ 1, spkrac == 1 ~ 0),

    colrac_r = case_when(colrac == 5 ~ 1, colrac == 4 ~ 0),

    librac_r = case_when(librac == 1 ~ 1, librac == 2 ~ 0),
  )

```

```

#Recode the homosexuality data as either tolerant or intolerant
new_filtered_data2homo <- new_filtered_data2homo |>
  mutate(

    spkhomo_r = case_when(spkhomo == 2 ~ 1, spkhomo == 1 ~ 0),

    colhomo_r = case_when(colhomo == 5 ~ 1, colhomo == 4 ~ 0),

    libhomo_r = case_when(libhomo == 1 ~ 1, libhomo == 2 ~ 0)
  )

```

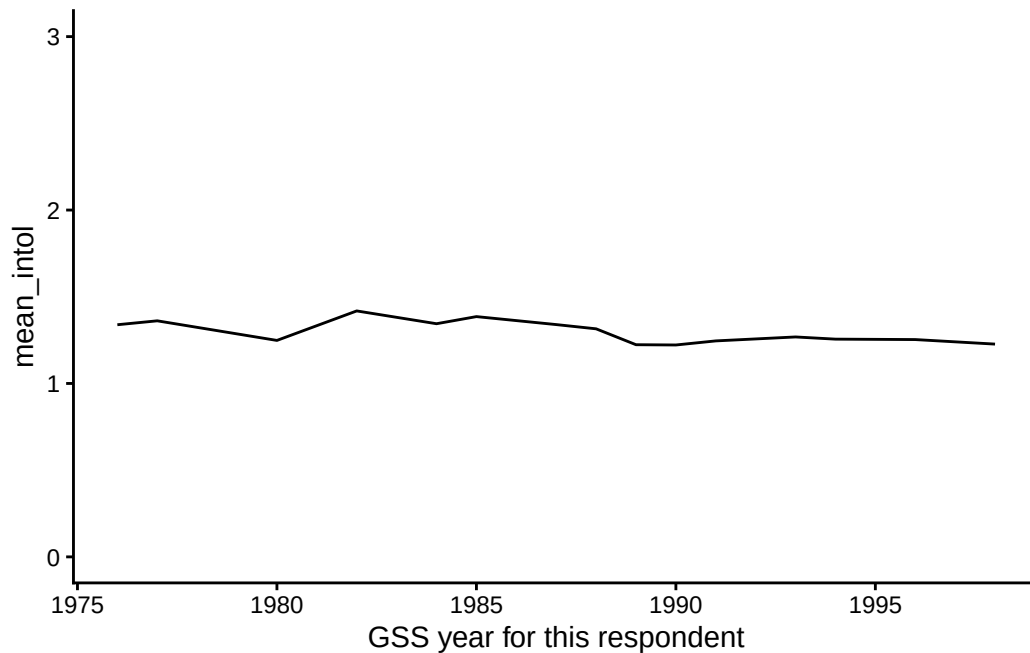
```

#Regroup the recoded and filtered racist data by year. Create a line graph
new_filtered_data2rac <- new_filtered_data2rac |>
  mutate(intol = rowSums(across(ends_with("rac_r")))) |>
  filter(!is.na(intol))

yearly_rac <- new_filtered_data2rac |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly_rac, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 3)) +
  theme_classic()

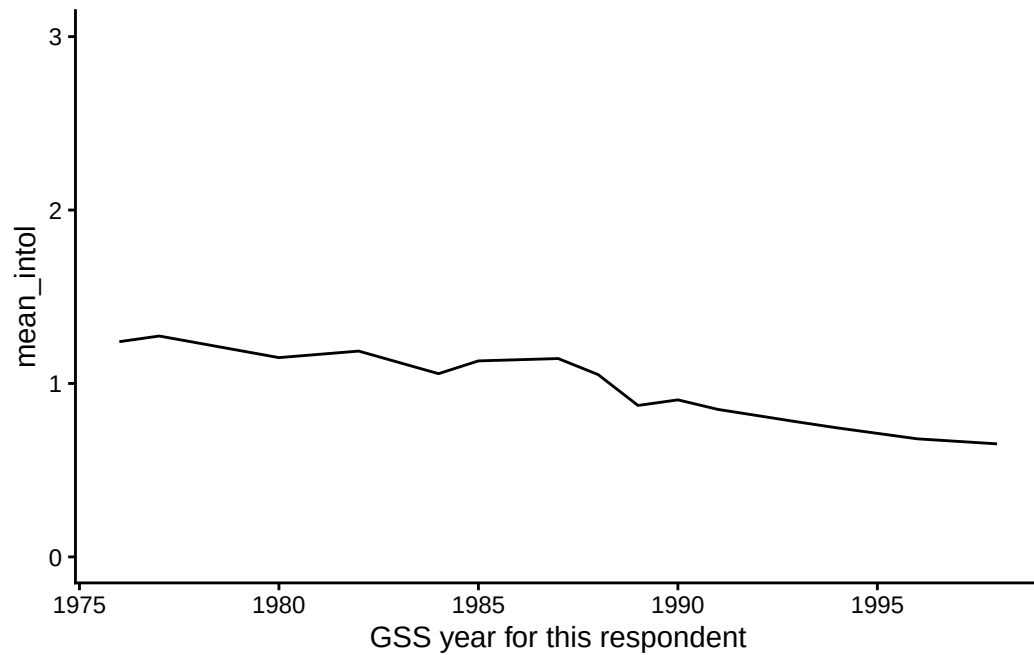
```



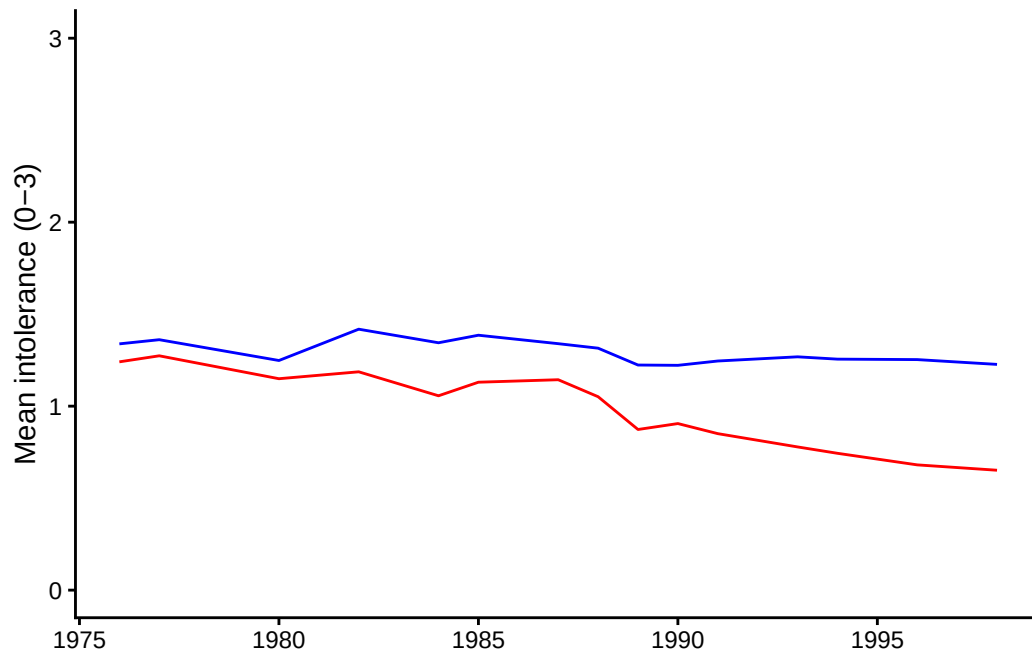
```
##Regroup the recoded and filtered homosexuality data by year. Create a line graph
new_filtered_data2homo <- new_filtered_data2homo |>
  mutate(intol = rowSums(across(ends_with("homo_r")))) |>
  filter(!is.na(intol))

yearly_homo <- new_filtered_data2homo |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly_homo, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 3)) +
  theme_classic()
```

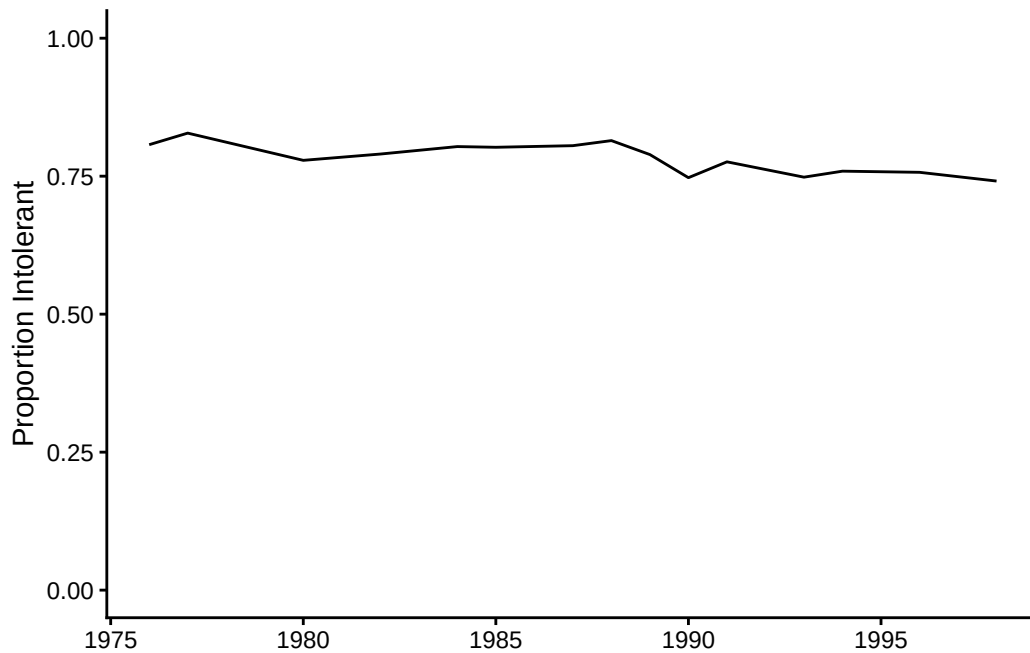


```
#Create a graph that combines the two
ggplot() +
  geom_line(data = yearly_rac, aes(x = year, y = mean_intol), color = "blue") +
  geom_line(data = yearly_homo, aes(x = year, y = mean_intol), color = "red") +
  scale_y_continuous(limits = c(0, 3)) +
  labs(x = NULL, y = "Mean intolerance (0-3)") +
  theme_classic()
```



```
#Create a graph that groups by year and shows the proportion of tolerance versus intolerance
yearly5 <- new_filtered_data |>
  mutate(intolerant = if_else(intol > 0, 1, 0)) |>
  group_by(year) |>
  summarise(prop_intol = mean(intolerant))

ggplot(yearly5, aes(x = year, y = prop_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 1)) +
  labs(x = NULL, y = "Proportion Intolerant") +
  theme_classic()
```

```
#Recode all filtered data as either tolerant or intolerant
filtered_data <- filtered_data |>
  mutate(

    spkath_r = case_when(spkath == 2 ~ 1, spkath == 1 ~ 0),
    spkrac_r = case_when(spkrac == 2 ~ 1, spkrac == 1 ~ 0),
    spkcom_r = case_when(spkcom == 2 ~ 1, spkcom == 1 ~ 0),
    spkmil_r = case_when(spkmil == 2 ~ 1, spkmil == 1 ~ 0),
    spkhomo_r = case_when(spkhomo == 2 ~ 1, spkhomo == 1 ~ 0),

    colath_r = case_when(colath == 5 ~ 1, colath == 4 ~ 0),
    colrac_r = case_when(colrac == 5 ~ 1, colrac == 4 ~ 0),
    colmil_r = case_when(colmil == 5 ~ 1, colmil == 4 ~ 0),
    colhomo_r = case_when(colhomo == 5 ~ 1, colhomo == 4 ~ 0),

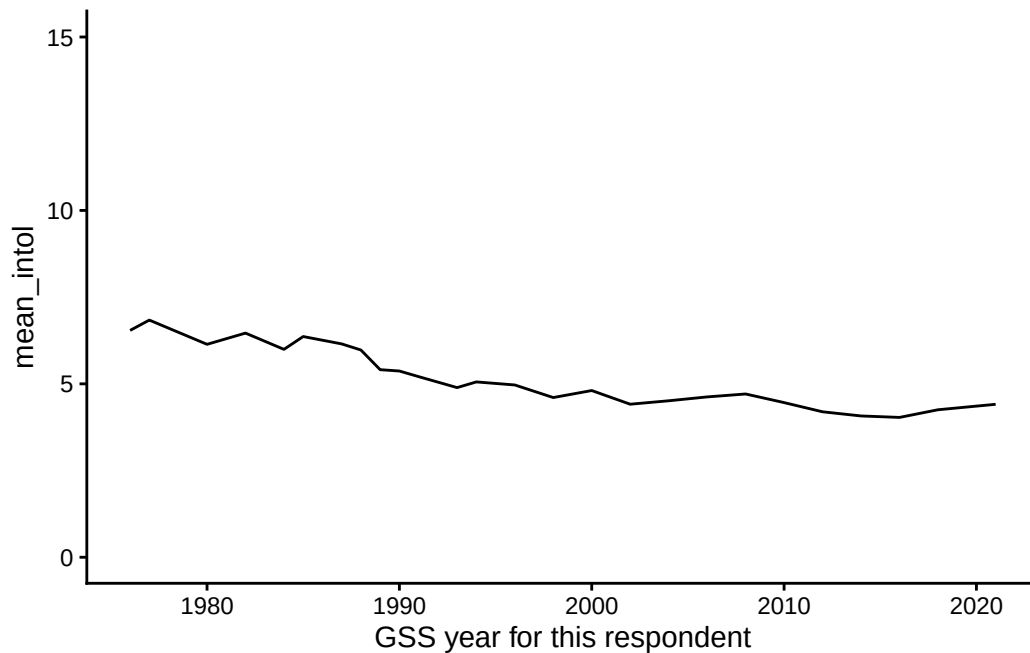
    colcom_r = case_when(colcom == 4 ~ 1, colcom == 5 ~ 0),

    libath_r = case_when(libath == 1 ~ 1, libath == 2 ~ 0),
    librac_r = case_when(librac == 1 ~ 1, librac == 2 ~ 0),
    libcom_r = case_when(libcom == 1 ~ 1, libcom == 2 ~ 0),
    libmil_r = case_when(libmil == 1 ~ 1, libmil == 2 ~ 0),
    libhomo_r = case_when(libhomo == 1 ~ 1, libhomo == 2 ~ 0)
  )
```

```
#Group by year and create a graph without year filtering
filtered_data <- filtered_data |>
  mutate(intol = rowSums(across(ends_with("_r")))) |>
  filter(!is.na(intol))

yearly <- filtered_data |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 15)) +
  theme_classic()
```



```
#Recode all racist filtered data as either tolerant or intolerant
filtered_data2rac <- filtered_data2rac |>
  mutate(

    spkrac_r = case_when(spkrac == 2 ~ 1, spkrac == 1 ~ 0),

    colrac_r = case_when(colrac == 5 ~ 1, colrac == 4 ~ 0),
```

```

  librac_r = case_when(librac == 1 ~ 1, librac == 2 ~ 0),
)

```

```

#Recode all homosexuality filtered data as either tolerant or intolerant
filtered_data2homo <- filtered_data2homo |>
  mutate(

    spkhomo_r = case_when(spkhomo == 2 ~ 1, spkhomo == 1 ~ 0),

    colhomo_r = case_when(colhomo == 5 ~ 1, colhomo == 4 ~ 0),

    libhomo_r = case_when(libhomo == 1 ~ 1, libhomo == 2 ~ 0)
  )

```

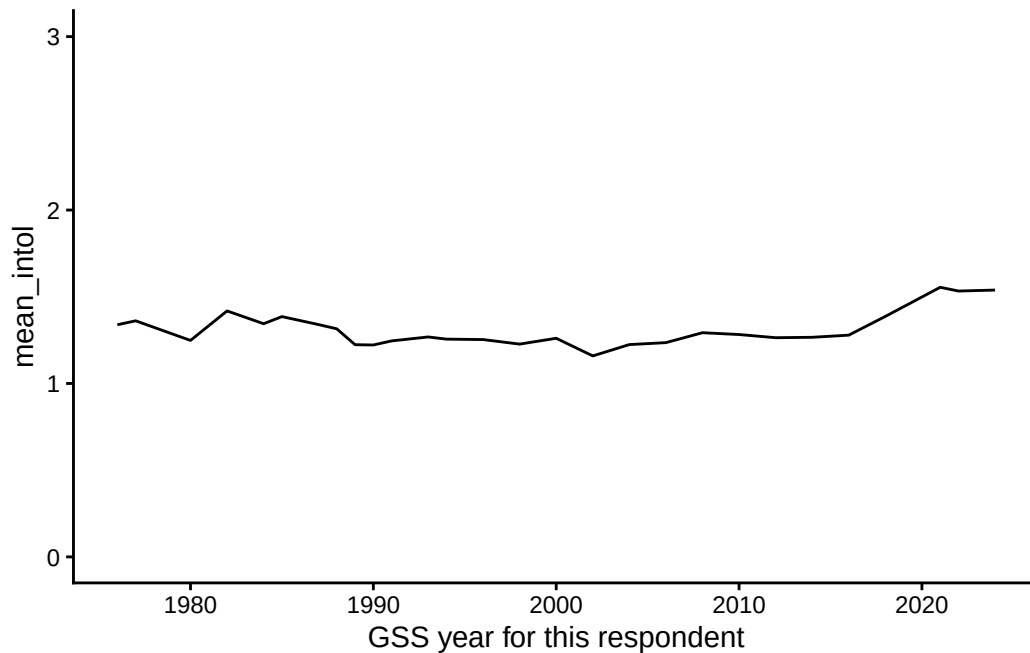
```

#Create the same racist line graph as above but without filtering for year and using all fil
filtered_data2rac <- filtered_data2rac |>
  mutate(intol = rowSums(across(ends_with("rac_r")))) |>
  filter(!is.na(intol))

yearly_rac <- filtered_data2rac |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly_rac, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 3)) +
  theme_classic()

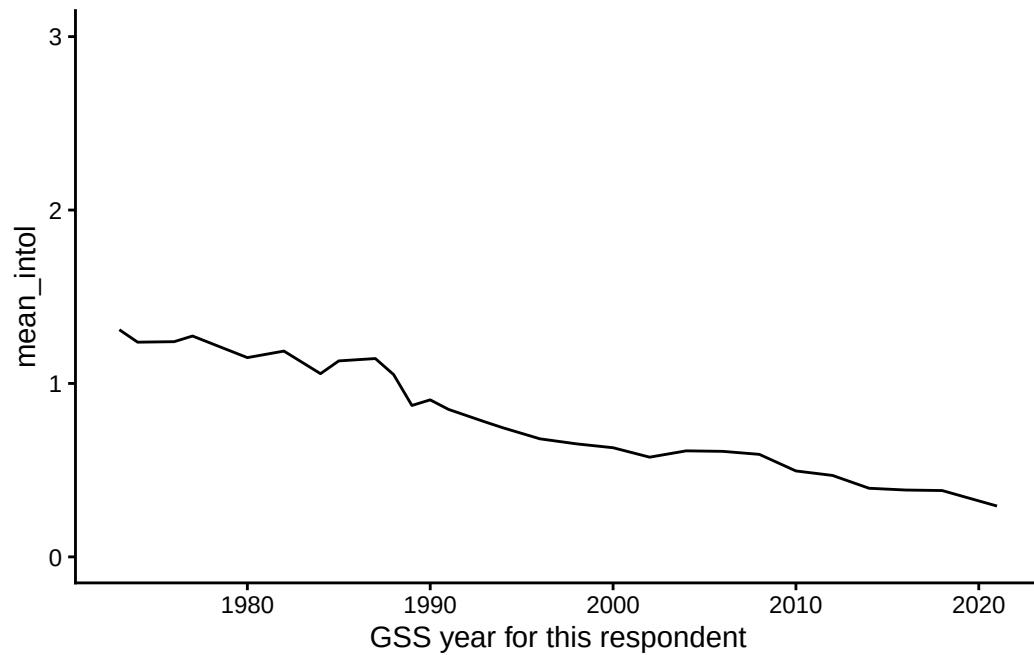
```



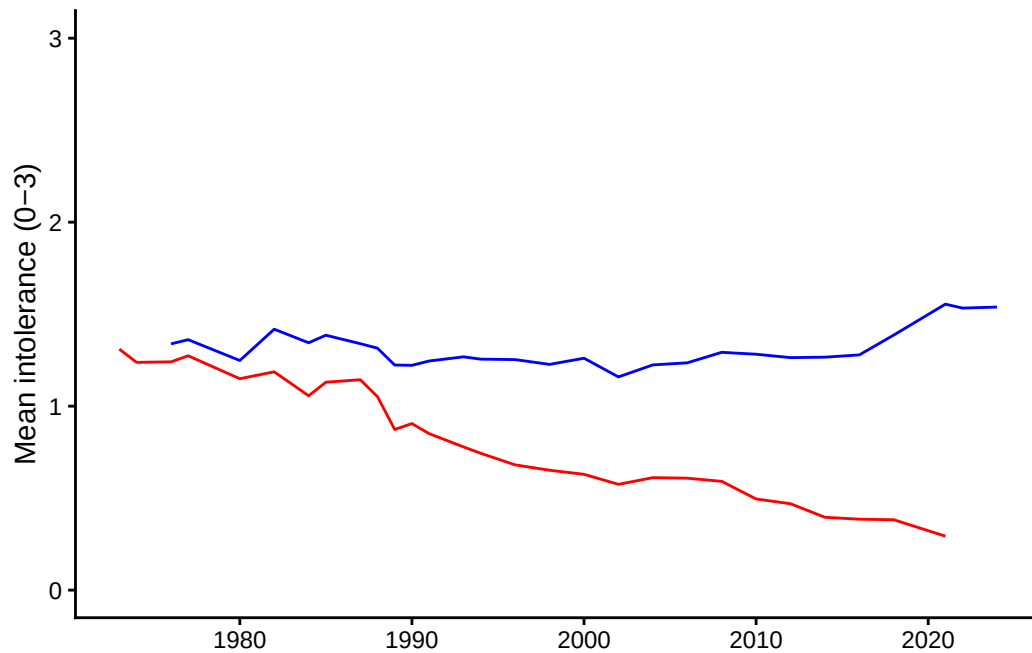
```
#Create the same homosexuality line graph as above but without filtering for year and using a
filtered_data2homo <- filtered_data2homo |>
  mutate(intol = rowSums(across(ends_with("homo_r")))) |>
  filter(!is.na(intol))

yearly_homo <- filtered_data2homo |>
  group_by(year) |>
  summarise(mean_intol = mean(intol))

ggplot(yearly_homo, aes(x = year, y = mean_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 3)) +
  theme_classic()
```

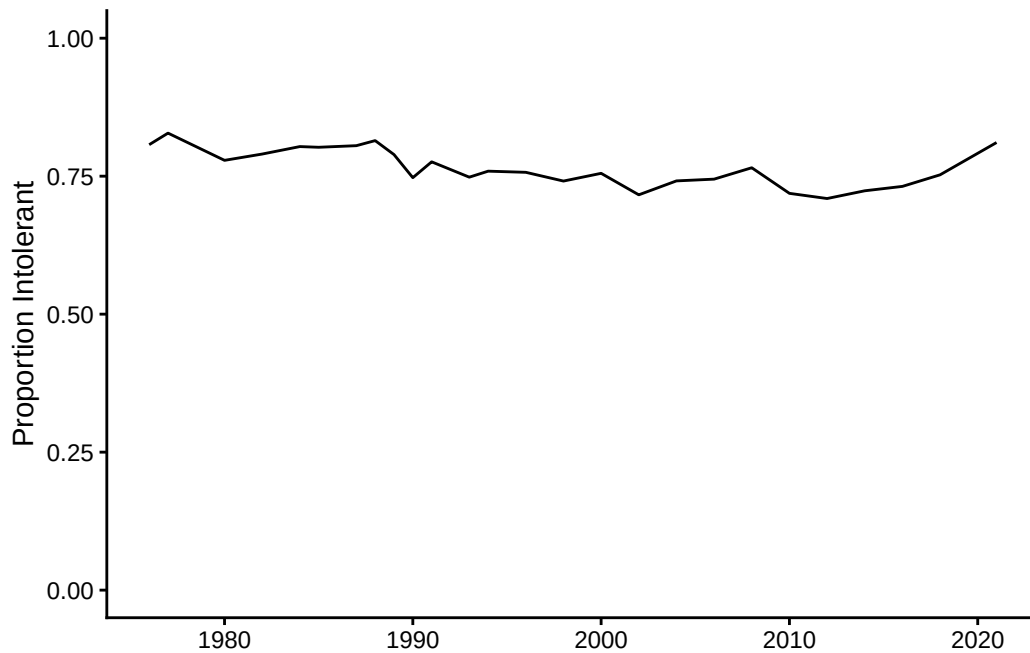


```
#Create a graph that combines the two
ggplot() +
  geom_line(data = yearly_rac, aes(x = year, y = mean_intol), color = "blue") +
  geom_line(data = yearly_homo, aes(x = year, y = mean_intol), color = "red") +
  scale_y_continuous(limits = c(0, 3)) +
  labs(x = NULL, y = "Mean intolerance (0-3)") +
  theme_classic()
```



```
#Create a proportion intolerance graph, but with all filtered data
yearly5 <- filtered_data |>
  mutate(intolerant = if_else(intol > 0, 1, 0)) |>
  group_by(year) |>
  summarise(prop_intol = mean(intolerant))

ggplot(yearly5, aes(x = year, y = prop_intol)) +
  geom_line() +
  scale_y_continuous(limits = c(0, 1)) +
  labs(x = NULL, y = "Proportion Intolerant") +
  theme_classic()
```



```
#Set aside new variables and make them into a new data set
compelvars <- c("year", "helpfulv", "trustv")
compelfiltereddata <- mydata[, compelvars]
```

```
#Account for NA answers
compelfiltereddata <- compelfiltereddata |>
  mutate(
    helpfulv_f = as_factor(helpfulv),
    helpfulv_f = fct_na_value_to_level(helpfulv_f, level = "NA")
  )
```

```
#Recode and create a line graph showing the proportion of people who identify as helpful or t
compel <- compelfiltereddata |>
  mutate(
    helpful_r = case_when(helpfulv == 1 ~ 1,
                          helpfulv %in% 2:3 ~ 0),
    trust_r   = case_when(trustv == 1 ~ 1,
                          trustv %in% 2:3 ~ 0)
  )

yearly_compel <- compel |>
  group_by(year) |>
  summarise(
```

```

    Helpful = mean(helpful_r, na.rm = TRUE),
    Trust    = mean(trust_r,   na.rm = TRUE)
  ) |>
  pivot_longer(-year, names_to = "measure", values_to = "proportion") |>
  filter(!is.na(proportion))

ggplot(yearly_compel, aes(x = year, y = proportion, linetype = measure)) +
  geom_line() +
  scale_y_continuous(limits = c(0, .5)) +
  labs(x = NULL, y = "Proportion", linetype = NULL) +
  theme_classic()

```

